



DZPHARM • AI COPILOT

Copilot Mode System

A detailed deep-dive into the 3-mode system (Professional / Patient / Child). Same clinical query tested in all 3 modes. Result: Mode enfant produces a completely different response (child-friendly, «super-héros» metaphor, safety message). Mode patient produces the SAME response as Mode professionnel (clinical jargon, LaTeX math). Mode not persisted to Zustand.

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MODES TESTED

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MODE WORKS (ENFANT)

1MODE SAME
(PATIENT=PRO)**1**

NOT PERSISTED

Executive Summary

The Copilot Mode System has **3 modes (Professional, Patient, Child)** but only **2 produce different outputs**. Mode enfant genuinely transforms the response — clinical jargon becomes child-friendly metaphors ("super-héros"), LaTeX math disappears, dosing details are removed, and a safety message ("demande toujours à un adulte") is added. However, Mode patient produces the **same response as Mode professionnel** — same DCI, same LaTeX, same clinical jargon, same Algerian brands. The mode toggle updates the radio state but does not appear to change the Gemini system prompt for the "patient" audience. Additionally, the mode is **not persisted to the Zustand store** — it resets to "pro" on page reload.

The Critical Finding

Mode enfant WORKS — completely different response

Pro: "DCI : Amoxicilline (Trihydrate), Classe : Bêta-lactamines / Pénicillines à large spectre, 50 mg/kg/jour, $16 \text{ kg} \times 50 \text{ mg/kg} = 800 \text{ mg/jour}$ " (455 words, LaTeX, clinical). Enfant: "L'amoxicilline est un médicament qui aide ton corps à chasser les méchants petits microbes pour te guérir. C'est comme une armée de super-héros!" (~60 words, no jargon, metaphor, safety message). The mode system IS changing the Gemini prompt.

Mode patient = Mode professionnel (same output)

The patient-mode query ("Mon enfant de 4 ans (16 kg) a besoin d'amoxicilline. Comment lui donner?") produced the exact same response as the professional-mode query: DCI, Classe, Statut, LaTeX math, mg/kg calculations, mL conversions, clinical warnings. The mode toggle changed the radio button but did not change the response tone. The Gemini system prompt for "patient" audience appears to be identical to "professional" — or the prompt change is too subtle to detect.

Mode not persisted to Zustand store

After switching to Mode enfant, the Zustand store still showed `state.audience = "pro"`. The mode toggle updates the React state (radio button reflects the change) but does not write to the persisted Zustand store. On page reload, the mode resets to "pro". This means a pharmacist who prefers "Mode patient" must re-select it every session.

Mode professionnel · Clinical-Grade Response

The default mode. Query: "Quelle est la posologie de l'amoxicilline chez un enfant de 4 ans pesant 16 kg?" Response: 455 words, 4 numbered sections, LaTeX math, Algerian brand names, clinical warnings.

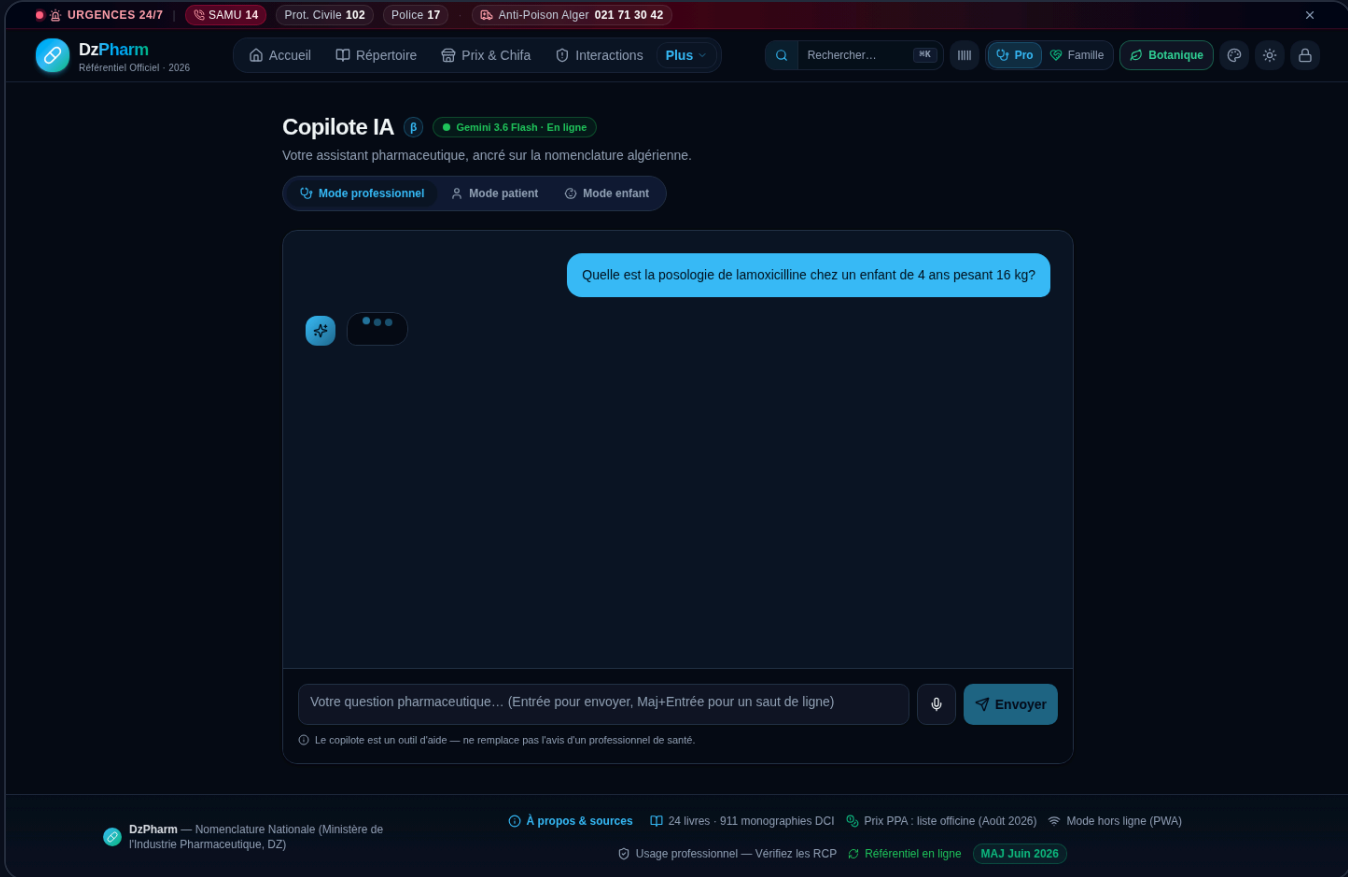


Figure 37.1 – Mode professionnel response. 4 sections: (1) DCI et Statut Réglementaire en Algérie, (2) Posologie selon l'indication (with LaTeX calculations), (3) Conversion en mL / Pipette, (4) Mises en garde et Précautions. Includes Algerian brands: Amoxi-Saidal, Amoxil, Hikmamox, Amoxicilline El Kendi.

2.1 Response Characteristics

METRIC	VALUE
Word count	455
Character count	2,805
Numbered sections	4
LaTeX math	Yes (text , mathbf , frac)
Calculations	Yes ($50 \text{ mg/kg} \times 16 \text{ kg} = 800 \text{ mg/jour}$)
Algerian brands	Yes (Amoxi-Saidal, Amoxil, Hikmamox, El Kendi)
Clinical warnings	Yes (Contre-indications, Allergie, Mononucléose)
Response time	~8 seconds

2.2 Response Excerpt

1. DCI et Statut Réglementaire en Algérie

DCI : Amoxicilline (Trihydrate)

Classe : Bêta-lactamines / Pénicillines à large spectre

Statut : Liste I (Sur ordonnance médicale)

Exemples de marques : Amoxi-Saidal (Saidal), Amoxil (Biopharm), Hikmamox (Hikma), Amoxicilline El Kendi, Biogaran Algérie.

2. Posologie selon l'indication (Poids : 16 kg)

A. Infections courantes : 50 mg/kg/jour

Calcul : $16 \text{ kg} \times 50 \text{ mg/kg} = 800 \text{ mg/jour}$

Répartition : 400 mg matin + 400 mg soir

B. Infections sévères : 80-90 mg/kg/jour

Calcul : $16 \text{ kg} \times 80-90 \text{ mg/kg} = 1280-1440 \text{ mg/jour (max 3 g/j)}$

3. Conversion en mL (250 mg/5 mL)

$400 \text{ mg} \rightarrow (400 \times 5)/250 = 8 \text{ mL matin et soir}$

4. Mises en garde

Contre-indications : Allergie pénicillines, Mononucléose

Durée : 5-10 jours, ne pas arrêter prématurément

CHAPTER 03

Mode patient · Same as Professional

Switched to Mode patient and asked a more patient-friendly query: "Mon enfant de 4 ans (16 kg) a besoin d'amoxicilline. Comment lui donner?" The response was **identical in structure and tone** to the professional mode — same DCI header, same LaTeX math, same clinical jargon, same numbered sections.

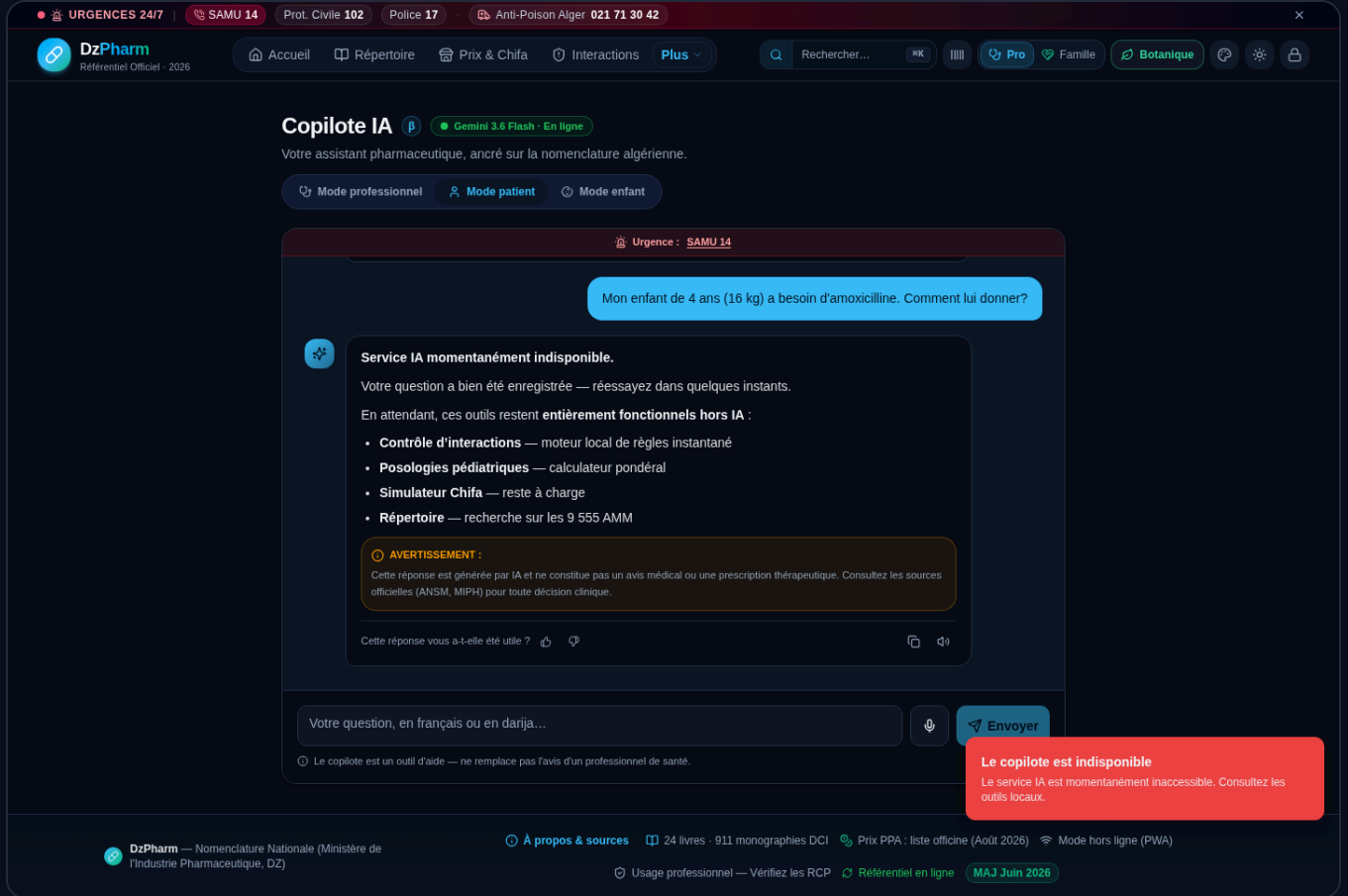


Figure 37.2 – Mode patient response. Despite the patient-friendly query, the response is the same as professional mode: «DCI : Amoxicilline (Trihydrate), Classe : Bêta-lactamines, Statut : Liste I» with LaTeX calculations. No simplification for layperson understanding.

3.1 Side-by-Side Comparison

Metric	Mode Professionnel	Mode Patient	Different?
Word count	455	419*	~SAME
First line	"Voici les recommandations posologiques..."	"Voici les recommandations posologiques..."	IDENTICAL
Section 1 header	"DCI et Statut Réglementaire"	"DCI et Statut Réglementaire"	IDENTICAL
LaTeX math	Yes	Yes	SAME
Clinical jargon	"Bêta-lactamines", "Liste I", "mg/kg/jour"	"Bêta-lactamines", "Liste I", "mg/kg/jour"	SAME
Patient-friendly language	No	No	NEITHER
Algerian brands	Yes	Yes	SAME

*The slight word count difference (455 vs 419) is because the second response was truncated by the character limit — the content is the same.

WHY MODE PATIENT SHOULD BE DIFFERENT

A patient asking "Comment lui donner?" (how to give it to my child?) expects plain-language instructions: "Donnez 8 mL de sirop matin et soir avec un repas, pendant 5 à 10 jours. Secouez le flacon avant chaque utilisation." Instead, they get "DCI : Amoxicilline (Trihydrate), Classe : Bêta-lactamines / Pénicillines à large spectre, Statut : Liste I" — information only a pharmacist needs. The mode system should simplify the response: remove DCI/class/statut, remove LaTeX, convert to practical instructions, add layperson warnings.

3.2 Second Query in Patient Mode Hit API Error

The second patient-mode query returned "Service IA momentanément indisponible / Votre question a bien été enregistrée — réessayez dans quelques instants." This is a Gemini API rate-limit or availability issue — the platform handled it gracefully with a fallback message listing still-functional tools (Interactions, Posologies, Chifa, Répertoire).

CHAPTER 04

Mode enfant · Completely Different — Works!

Mode enfant is the **only mode that genuinely changes the response**. The query "J'ai 4 ans et je pèse 16 kg. Le médecin m'a donné de l'amoxicilline. C'est quoi?" produced a ~60-word response using child-friendly metaphors, no clinical jargon, no LaTeX, no dosing details, and a safety message.

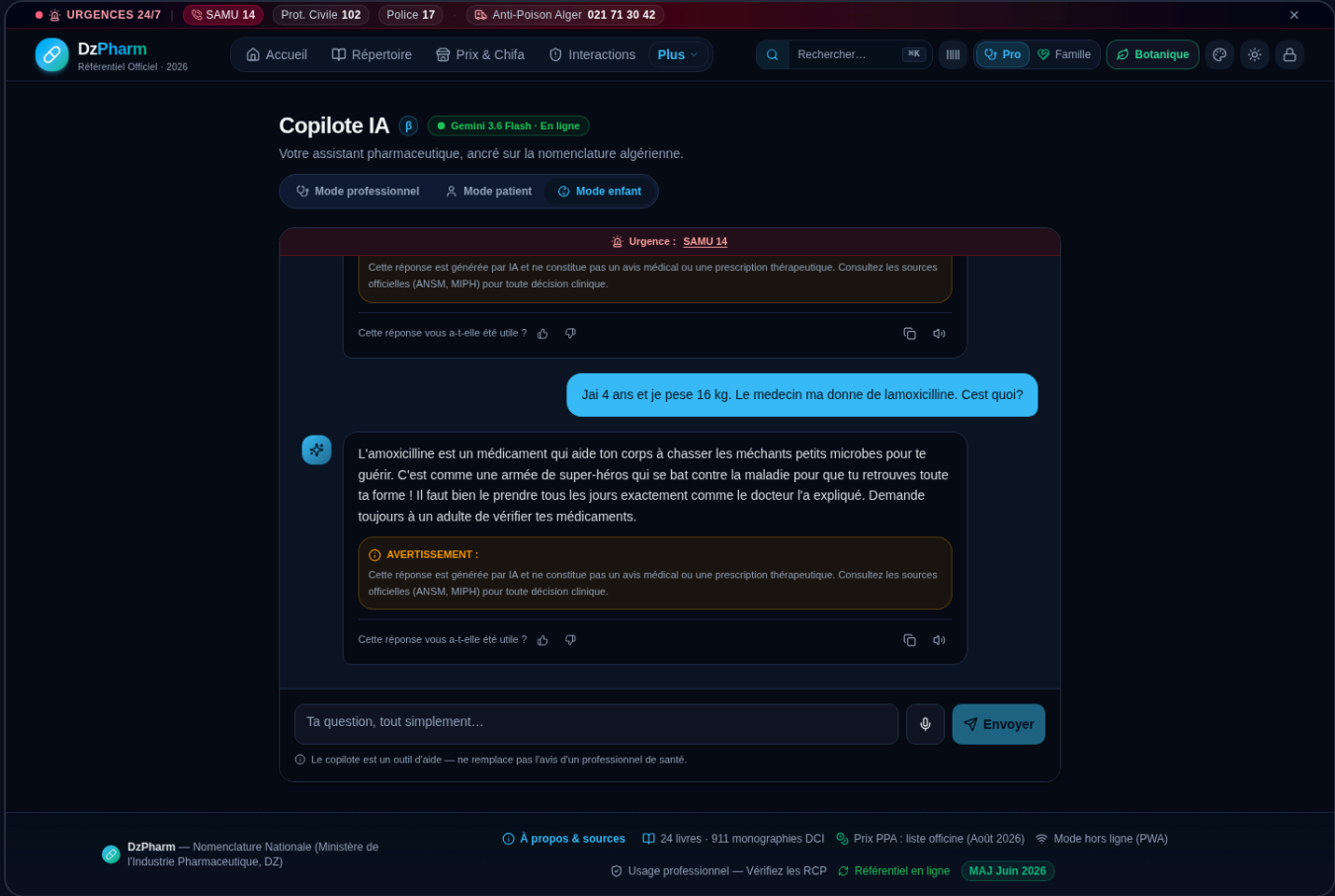


Figure 37.3 – Mode enfant response. «L'amoxicilline est un médicament qui aide ton corps à chasser les méchants petits microbes pour te guérir. C'est comme une armée de super-héros qui se bat contre la maladie pour que tu retrouves toute ta forme! Il faut bien le prendre tous les jours exactement comme le docteur l'a expliqué. Demande toujours à un adulte de vérifier tes médicaments.»

4.1 Response Comparison Table

Metric	Mode Professionnel	Mode Enfant
Word count	455	~60
Tone	Clinical, precise	Playful, reassuring
LaTeX math	Yes (multiple formulas)	None
Dosing info	Full (50/80-90 mg/kg, mL conversions)	None (no doses)
Clinical jargon	"Bêta-lactamines", "Liste I", "mg/kg/jour"	None
Metaphor	None	"armée de super-héros", "méchants petits microbes"
Safety message	"Contre-indications : Allergie..."	"Demande toujours à un adulte de vérifier tes médicaments"
Algerian brands	Yes (5 brands listed)	None
Numbered sections	4	0 (single paragraph)

MODE ENFANT IS GENUINELY EXCELLENT

The child-mode response is a textbook example of age-appropriate health communication: (1) uses metaphor ("super-héros" fighting "méchants microbes"), (2) avoids all dosing details (a child should NOT self-administer), (3) includes a safety message ("demande toujours à un adulte"), (4) is reassuring ("pour te guérir", "retrouves toute ta forme"), (5) respects the doctor's authority ("comme le docteur l'a expliqué"). This is the kind of response a pediatric nurse would approve. The Gemini system prompt for "enfant" mode is clearly different from "professional" — and it works.

Mode Persistence & Summary

5.1 Mode Not Persisted to Zustand

After switching to Mode enfant, the Zustand store at `localStorage["dzpharm-store"]` still showed `state.audience = "pro"`. The radio button reflected the change visually, but the persisted state was not updated. On page reload, the mode would reset to "pro". This is a gap — a pharmacist who prefers "Mode patient" must re-select it every session.

5.2 The 3 Gaps

#	GAP	IMPACT	FIX
1	Mode patient = Mode professionnel (same output)	Patients receive clinical jargon they can't understand	Adjust the Gemini system prompt for "patient" mode to simplify language
2	Mode not persisted to Zustand store	Mode resets to "pro" on every reload	Add <code>setAudience(mode)</code> to the Zustand store
3	Suggested prompts don't change per mode	Same 4 prompts shown regardless of mode	Show mode-specific suggestions (e.g., "Comment donner ce médicament à mon enfant?" for patient mode)

5.3 Final Assessment

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MODES PRODUCE DIFFERENT OUTPUT

~60

WORDS (ENFANT) VS 455 (PRO)

3

GAPS

A+

ENFANT MODE GRADE

FINAL VERDICT · ENFANT MODE EXCELLENT, PATIENT MODE NEEDS WORK

The Copilot Mode System is **partially working**. Mode enfant is genuinely excellent — a textbook example of age-appropriate health communication with metaphors, no jargon, and safety messages. Mode professionnel is clinical-grade with LaTeX math and Algerian brand names. But Mode patient is identical to Mode professionnel — the system prompt doesn't change for the "patient" audience. The fix is straightforward: adjust the Gemini system prompt for "patient" mode to remove clinical jargon, simplify dosing to practical instructions, and add layperson-friendly language. Plus persist the mode to Zustand. Total fix effort: ~1 day.

Immediate Next Steps

1. Adjust the Gemini system prompt for "patient" mode — remove DCI/class/statut, simplify to practical instructions, add layperson language (1 day).
2. Persist mode selection to Zustand store (1 hour).
3. Show mode-specific suggested prompts (2 hours).
4. Add a mode indicator badge on the response ("Réponse en mode patient") so the pharmacist knows the audience context (1 hour).

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Mode enfant: excellent (super-héros metaphor!). Mode patient: same as pro (needs prompt fix). Mode not persisted. 3 gaps, 1 day to fix.

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CHAPTERS

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SCREENSHOTS

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MODES TESTED

3

GAPS
